

stylish speaker sets we received, but there was some minor shielding issue with it.

TECH-COM's SSD-2800 is a pseudo-4.1 set—which means the rear speakers are practically just playing the same stuff as the fronts. To make matters worse, the cables are extremely short, so you can only place them right next to the front speakers. The SSD-3400 from TECH-COM was another such speaker. While these may have four satellite speakers, they're still good for stereo.

The Logitech G51 comes with thick cables, colour-coded to make it extremely easy to set up. Even the Altec Lansing FX5051 was considerably easy to set up for the same reason. These are examples of attention to detail, along with detailed troubleshooting information that make up a good set of speakers.

Some of the TECH-COM and Intex speakers, like those in the 2.1 category, had some very attractive designs.

There were some problems we faced while trying to get proper surround sound from some speakers, though. On some of the TECH-COMs, we had to plug in the front channel tracks



Logitech Z-5500

into the auxiliary port to get 2.1 sound. Another general issue with most TECH-COM speakers is the lack of 3.5mm-RCA converters. There were also very severe shielding issues seen in the TECH-COM SSD-5001R and the Intex IT-4800. Placing the centre speaker over a CRT monitor—or anywhere near it—is out of the question.

Fooobar Frequency Test

Very low deep bass along with power is necessary for a satisfying movie experience—for bomb blasts and crazy dogfights, among

other things. The Z-5500's level of bass at 20 Hz is pretty low as compared to the other speakers—it's the sheer size of the driver that could be to blame here.

We also saw very erratic performance from all the other speakers. The Altec Lansing FX5051, for example, is extremely loud on higher frequencies, and not so much on the mids and lower range.

The sole Supercomp in the surround speaker category—the SES 5177—performed abysmally. The woofer was all over the place in the low frequency test. There was no clear steady tone but randomly switching tone. The Creative Inspire M5300 also lacked presence of bass between the 20-50 Hz frequency ranges but it was still a fair bit better than the M4500.

Music Test

Logitech's Z-5500 seemed to be doing really well in most of the tests, and we assumed it would do equally well at music with its fairly neutral tone. It performs quite respectably, but bass isn't as tight as the Altec Lansing MX5021 from the 2.1 speaker category. The double bass drumming in the Metal tracks didn't sound as pronounced as we'd like. For Bollywood pop music, the punchy bass did well. The Z-5500s are also very loud, so you can actually have your friends over for your very own party in an average-sized room.

The Logitech G51 has a heavy but underpowered woofer. It gives you good soft bass in tracks like *With or without you* by U2. The satellites are pretty

decent, and the sound is rather warm and neutral without any shrill treble. The remote has a "matrix mode" for gaming and music, and fares horribly. The bass completely disappeared, so it's wise to keep the effects off at all times.

The Altec Lansing VS3251 isn't powerful, but the quality and detailed reproduction of sound was pretty impressive regardless. The TECH-COM SSD-4301R was surprisingly decent—better than most of the other TECH-COMs. It wasn't as crystal clear or crisp as the Altec Lansings, but the woofer pumped out well-balanced bass and it was pretty powerful as well. Another TECH-COM that did well was the SSD-5101R. The bass was very good for drum solos—both in the quality tests and in the drum solo by Mike Portnoy in the live version of Dream Theatre's *Beyond this life*.

The Philips MMS460 surprised us quite a bit—it has a very light woofer, but it gives some really impressive bass that does justice to bass guitars. It lacks the punchy hard bass for the drums, though, and the satellites exaggerate the higher frequencies. Bollywood and pop music was given an extra sparkle on top range frequencies, which made for a unique sound. The Philips MMS460 sound pretty nice in the piano tracks with every single note pumping that essential little bit of bass. Similar sound characteristics could be heard in the pure vocal (*a capella*) tracks with finger snapping.

Movies

The Logitech Z-5500 does a fair job at movies. The ample power makes the action-packed *Star Wars* and *Transformers* clips a joy to go through. The Z-5500s are rated at 500 W RMS, but it really doesn't sound that much louder than the Logitech MX5021—which is around 90 W RMS.

We're pretty impressed by the Logitech M-50 in the music tests, even though they are low-power speakers. Movies sound all right, but for quiet movie scenes, the bass wouldn't play its part at all. Although not very loud, the detail was still very respectable. They are rated at 26 W RMS, but sound pretty powerful.



Altec Lansing VS3251

The Altec Lansing FX5051 gives extremely good detail in the *Star Wars* scene—the quality of the crunch of a ship smashing into another was pretty amazing. The lack of punch in the woofer and the distortion at higher volumes was a complete turn-off. The woofer had to be turned up to over 80 per cent to get some bass into the sound.

Extras

The Logitech Z-5500 comes with a remote control with a display on it. Along with the remote is an inbuilt decoder. All the features and processing make the remote panel a little hot after some use. The remote itself is feature-packed with different audio modes for listening to music and movies, or using all speakers for music.

Yg Acoustics Voyager



The Yg Acoustic Voyager's distinctive design looks like something out of a sci-fi movie. Unlike all the other mammoths of its class, the weight of the Voyager is just around 91 Kg thanks to use of

cast aluminium. The price for the Voyager is a comparatively low \$1,00,000. The Voyager is so exotic that it is only sold to the well known in the music and film industry. No, you can't walk in for a demonstration—you are to be invited by them.

The FX5051 can also be driven through a USB interface if you don't have sound card. There were no problems faced with detection, or anything to do with drivers. The quality is nothing special either, and these days pretty much every single motherboard comes with integrated sound so it doesn't do a lot.

The Best!

We recommend the Z-5500 for its good quality, and more importantly, its colossal power. The biggest problem with the Z-5500 is the price—Rs 27,995! For that price, you don't really get that many more times the power or quality that you would get from a very good 2.1 like the Altec Lansing MX5021.

If power isn't what you're looking for, then the Altec Lansing VS3251 is a good set which is very affordable and is sufficient for an audience of two or three people in front of your PC. If you're willing to ignore the slightly high treble sound, then the Philips MMS460 is also

Room acoustics—why bother?

In the audiophile's bible, anything that reflects sound is evil. Well, most of the time, at least. Sound bounces off flat and hard surfaces and all you get from that is loads of stray echoes all over the place and some sounds end up overpowering others.

There are a few things you can do to improve the acoustics of your room. Firstly, you'll need to use some sound-

absorbing material on your walls and floors, and if possible, even your ceiling. Curtains, wooden shelves and stacked books also work as great absorbing materials. This way, sounds get absorbed rather than getting reflected. Moving the seating arrangement and positioning can also help you find a sweet spot where echoes are minimal. A very small room also has lots of resonance, and is a bad idea to set up a home theatre in such a place.

Some of the largest and most advanced anechoic chambers are used for the development and testing of aircraft and other vehicles for the armed forces. To give you an idea of the size, the largest anechoic (without echoes) chamber is the Benefield Anechoic Facility at the Edwards Air Force base in the US. The dimensions of the room are some 250 x 264 x 70ft, and fits 816,000 cones to snuff out reflected sound.

